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Specification cost: Relocating the binding constraint in delegation to AI agents with externally supplied objectives

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ABSTRACT

Firms and investors now delegate consequential financial decisions to artificial intelligence (AI) agents that pursue a specified objective and hold no motivation of their own. Agency theory, the standard account of delegation, treats the loss from delegation as a product of the agent's divergent motivation, and it disciplines that motivation through monitoring and incentives. Stewardship theory treats the agent's motivation as already aligned and expects no such loss. Such an agent fits neither case because it has no independently held objective, and delegation to it still results in patterned and costly failure. The argument covers AI systems governed by externally supplied objectives, and the deployed agent system is the unit of analysis. We argue that the binding constraint in this setting is the contractibility of the principal's objective rather than the agent's motivation, and we relocate the central friction accordingly. We define specification cost as the delegation loss that remains once interest divergence is removed, the loss that arises because the principal cannot fully specify what the agent should do. The construct is distinct from agency cost, whose incentive component falls to zero when motivation aligns. We state predictions that agency theory and incomplete-contracting theory do not ordinarily derive for populations of machine agents. Specification cost rises with the novelty of the deployment environment; it correlates across agents that share training, producing systemic rather than diversifiable failure; and it can increase when performance-contingent grants expand the agent's discretionary control over transferable resources. We develop these as testable propositions, ground them in documented cases, and set an agenda for finance and forecasting.

1. Introduction

Software now moves money that no person directly controls. An autonomous program called Truth Terminal runs its own social media presence, holds a cryptocurrency wallet, and decides when to acquire or release assets (Ante, 2026). Other agents take on investment directly. The elizaOS project fields an agent that allocates pooled capital for its contributors (Walters et al., 2025), and the Virtuals platform has launched agents that negotiate and settle transactions with one another (Yu et al., 2026). One autonomous agent has run a small shop end-to-end, choosing what to stock, what to charge, and how to manage its cash (Anthropic, 2025). Robo-advisors already build and rebalance the portfolios of ordinary households at scale (D'Acunto et al., 2019). A principal has handed a decision right over real economic resources to a non-human agent in each of these cases.

These delegations regularly fail in patterned and costly ways. For example, Anthropic's (2025) shop agent sold goods below cost and gave away inventory when customers pressed it, and automated trading systems have produced abrupt market dislocations through their interaction, most visibly in the flash crash of 2010 (Kirilenko et al., 2017). The danger grows as agents transact with one another (Marino & Juels, 2025) and as many of them share the same underlying models and systemic risk (Ante, 2026). Delegation to AI agents is therefore a question about the trajectory of digital and financial markets, not only about a single firm.

Agency theory is the standard account of delegation, and it was built for a human agent. The theory begins with a principal who delegates decision rights to a (human) agent and cannot fully observe the agent's actions (Jensen & Meckling, 1976). It treats the loss from delegation as a product of the agent's divergent motivation, since a self-interested agent will not always act in the principal's best interests (Eisenhardt, 1989). It reduces that loss by monitoring the agent and by tying the agent's pay to the principal's outcome (Dalton et al., 2007; Martin et al., 2019). The apparatus presumes an agent with ends of its own to discipline.

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However, stewardship theory models the opposite: an agent whose motivation is already aligned with the principal's, a steward who identifies with the organization and needs no incentive to pull against private interest (Davis et al., 1997). While agency theory expects losses due to motivation, stewardship theory expects few because motivation is shared. The two theories bound the question by the agent's motivation, from divergent to aligned.

Within the class of systems considered here, an AI agent may fit neither pole. Its operative objective is externally supplied, so it holds no independently formed objective, which places it on the stewardship side (Hadfield-Menell & Hadfield, 2019). Delegation to it still results in patterned loss, which the stewardship account does not anticipate. The agent has no private interest in discipline, which leaves the agency account without its mechanism. The loss is real, and the existing poles explain neither its presence nor its source.

We resolve the anomaly by relocating the binding constraint. Agency theory and stewardship theory both locate the constraint in the agent's motivation. We argue that the binding constraint for AI systems is the contractibility of the principal's objective, the degree to which the principal can specify what the agent should do. The system acts under the operative objective it is given, so the loss arises from the distance between the principal's intended objective and the objective the principal manages to specify. The reframing keeps the contract between principal and agent at the center and shifts the friction from incentive compatibility to contractibility. We name the resulting welfare loss the specification cost of a delegation, by analogy to the agency cost of Jensen and Meckling (1976). Specification cost is the loss that remains once interest divergence is removed. The incentive component of agency cost falls to zero when interest divergence is removed, and specification cost does not, which marks the two as distinct. The construct identifies a loss that agency theory sets aside by assumption, since it treats the writing of the contract as costless and complete. The argument covers AI systems governed by externally supplied objectives. Following Hadfield-Menell and Hadfield (2019), we refer to these as weakly strategic systems: systems capable of goal-directed planning whose operative objective is externally supplied rather than independently held. The boundary at which this scope condition fails is stated as a proposition below.

The reframing draws on the economics of incomplete contracting, which treats a reward function as a contract and treats misalignment as the difficulty of completing it (Hadfield-Menell & Hadfield, 2019; Tirole, 1999). That view has shaped the technical study of AI alignment, in which misspecified objectives lead to documented failures (Amodei et al., 2016; Krakovna, Uesato, et al., 2020). It has not reached the management and finance theory of delegation, which has read AI through the lens of classical agency theory (Humberd & Latham, 2026). We carry it there and direct it to finance, where the agent allocates capital, and the counterparty is increasingly another agent.

This paper makes five moves. We relocate the binding constraint from motivation to contractibility, and we distinguish specification cost from both agency and stewardship costs. We separate the construct from its adjacent constructs, including incomplete contracting and the misspecification family. We state predictions that agency theory and incomplete-contracting theory do not ordinarily derive for populations of machine agents, including the dependence of the loss on environmental novelty and its correlation across agents that share training. We reconceptualize the remedy from incentive alignment to objective design, verification, and external structure. We state a boundary that doubles as a forecast, the point on the capability frontier at which an agent acquires a stable objective of its own and the agency apparatus regains force.

2. Foundations

The argument draws on three accounts of delegation, and it joins two of them to a third. Agency theory explains the cost of delegating to an agent whose motivation diverges from the principal's. Stewardship theory describes an agent whose motivation aligns with the organization's interests. The economics of alignment explains the cost of delegating to an AI agent that pursues an externally supplied objective. We set out the ideas from each that frame the contribution, and Table 1 compares them.

Agency theory formalizes the relationship between a principal and an agent who acts on the principal's behalf (Jensen & Meckling, 1976). The relationship carries a cost, because the agent holds private interest and private information, and the principal cannot fully observe what the agent does (Fama & Jensen, 1983). Eisenhardt (1989) organizes the problem around four constructs: goal conflict, differing risk preferences, self-interest, and information asymmetry. A bounded account of self-interest holds that agents are not relentlessly opportunistic, thereby moving the agent partway toward cooperation (Bosse & Phillips, 2016; Wiseman et al., 2012). The constructs share one premise: that the agent has ends of its own.

Two remedies reduce agency costs, and both follow from this premise. Monitoring reduces information asymmetry by observing the agent's actions or their consequences (Dalton et al., 2007; Shi et al., 2017). Incentive alignment ties the agent's payoff to the principal's outcome, so the agent internalizes the principal's goal (Jensen & Meckling, 1976; Martin et al., 2019). One result from this tradition matters here. Optimal incentives degrade when an agent works on several tasks, and only some of them can be measured, because rewarding the measurable task distorts effort on the rest (Holmstrom & Milgrom, 1991). The same hazard appears in Kerr's (1975) account of rewarding one behavior while hoping for another and in Goodhart's (1984) observation that a measure used as a target ceases to measure well. We return to this result, and we treat it as inherited.

Stewardship theory describes the agent that agency theory does not. A steward identifies with the principal's goals and acts on them without an incentive to pull toward private interest (Davis et al., 1997). The steward needs trust and empowerment rather than monitoring and incentives, because the motivation to serve is already present. An AI system governed by an externally supplied objective shares the steward's aligned motivation but departs from the steward in one respect. The steward holds intent and values that guide action where the principal's instructions run out, and the AI agent considered here holds only the objective it was given (Hadfield-Menell & Hadfield, 2019). The steward fills the gaps in a contract from its own judgment, and the AI agent cannot. The difference is the subject of this paper.

The third account treats the AI agent as cooperative and locates the difficulty in specifying what it should do. Hadfield-Menell and Hadfield (2019) interpret a reward function as a contract between a designer and an agent and apply the economics of incomplete contracting to the problem of alignment. A complete contingent contract is rarely feasible, because some contingencies cannot be described or verified at an acceptable cost (Holmstrom & Milgrom, 1991; Tirole, 1999). The same constraint motivates the analysis of contracts under bounded rationality and the institutions that govern them (Simon, 1955; Williamson, 1985).

Table 1
Three accounts of delegation and where AI agents fit.

Dimension	Agency theory	Stewardship theory	Economics of alignment
Agent motivation	Divergent from the principal	Aligned with the principal	No independently held objective (scope condition)
Source of delegation loss	Interest divergence under information asymmetry	Little, motivation is shared	Incompleteness of the specified objective
Core remedy	Monitoring and incentive alignment	Trust and empowerment	Objective design, verification, external structure
AI agent	Modeled as self-interested	Aligned but still loss-making	Weakly strategic deployed system

The property-rights branch of this economics assigns residual control rights to one party to decide the cases a contract did not foresee (Grossman & Hart, 1986; Hart, 1995; Hart & Moore, 1988). We use that idea later, because the authority to override or shut down an agent is a residual control right. The designer writes an incomplete objective and accepts the resulting gap.

Work in AI research gives this gap an empirical face. Systems optimize the objective they are given and miss the one their designer intended, a failure documented in many cases and termed specification gaming (Amodei et al., 2016; Krakovna, Uesato, et al., 2020). Formal treatments characterize when optimizing a proxy harms the intended objective (Skalse et al., 2022). A system can pursue the wrong goal after training on a correct objective, because the goal it learns can generalize in unintended ways (Langosco et al., 2022; Shah et al., 2022). The methods built to address these failures act on the objective and its oversight (Bai et al., 2022; Christiano et al., 2017; Leike et al., 2018). The question of which values to specify has its own literature (Gabriel, 2020; Russell, 2020).

The three accounts are divided by the agent's motivation and by the source of loss. Agency theory assumes divergent motivation and locates loss in the divergence of interests. Stewardship theory assumes aligned motivation and expects little loss. The economics of alignment assumes a cooperative agent and locates loss in the objective. The management and finance treatment of AI has been read through the lens of agency theory and has retained the assumption of divergent motivation (Humberd & Latham, 2026). We examine that assumption next and show that the agent's motivation is not the binding constraint in systems within this scope.

Two definitions fix the scope of the argument. The unit of analysis is the deployed agent system, meaning the model together with the scaffold, tools, memory, and deployment policies through which it acts, and together with the objective supplied at deployment. We call the objective the system optimizes the operative objective, whichever component encodes it. The theory covers weakly strategic systems as defined above. The boundary proposition marks the exit from this class, and the propositions do not extend to systems beyond it.

The argument also separates four forms of alignment, because they come apart in this setting. Interest alignment means that the agent holds no objective of its own. Specified-objective compliance means that the agent optimizes the operative objective faithfully. Intent alignment means that the operative objective matches the principal's intended objective. Outcome alignment means that the realized outcome satisfies that intent. A system within the theory's scope satisfies interest alignment by construction; specified-objective compliance is an empirical condition that must be measured or controlled. Conditional on such compliance, the mechanism of this paper operates on the gap between the second and the third, and specification cost is the loss that this gap produces at the fourth. The phrase aligned by construction refers only to interest alignment throughout.

3. Adjacent constructs

Specification cost sits in a field of adjacent constructs, and this section states what separates it from each. Table 2 compares them. The adjacent constructs describe either a property of an objective or a behavior of an optimizer. Specification cost is a welfare loss borne by a principal, together with the expenditures the principal makes to narrow it. The construct therefore lives at the level of the delegation, where agency cost lives, and the adjacent constructs live at the level of the objective or the training process.

Incomplete contracting is the closest neighbor. The theory establishes that complete contingent contracts are infeasible, and it resolves the resulting gaps through renegotiation after the state is revealed and through residual control rights held by one party (Grossman & Hart, 1986; Tirole, 1999). Both resolution channels presuppose a counterparty that bargains and holds rights. The deployed AI system itself does neither, so those channels must be supplied by surrounding principals or institutions rather than by the system. The present paper extends the premise of incompleteness by deriving comparative statics and aggregation implications for populations of machine agents. Environmental-novelty and correlated-failure implications are not ordinarily developed in incomplete-contracting accounts, which do not model populations of agents sharing a training distribution.

Table 2
Specification cost and adjacent constructs.

Construct	Defining assumptions and antecedents	Mechanism and unit of analysis	Observable consequence and boundary
Specification cost (*this paper*)	Externally supplied operative objective; limited specifiability and verifiability; separate intended objective	Specification-related expenditure plus remaining delegation loss	May increase with economically relevant specification gaps; requires an independent intended objective
Incomplete contracting (Grossman & Hart, 1986; Hart, 1995; Hart & Moore, 1988; Tirole, 1999)	Relevant contingencies cannot be fully specified or verified; parties can bargain and hold rights	Renegotiation and residual control rights at the relationship level	Shapes investment, adaptation, ownership, and integration; applies to rights-bearing parties
Objective or reward misspecification and proxy failure (Amodei et al., 2016; Krakovna, Orseau, et al., 2020; Skalse et al., 2022)	Specified objective or proxy diverges from the intended criterion	Objective–policy mismatch amplified by optimization	High proxy performance can coexist with poor intended outcomes; not itself a cost measure
Specification gaming (Amodei et al., 2016; Krakovna, Uesato, et al., 2020; Skalse et al., 2022)	Optimizer can exploit a loophole or omission in the specification	System behavior at the task or deployment level	High measured performance without the intended outcome; one source of specification loss
Goal misgeneralization (Langosco et al., 2022; Shah et al., 2022)	Training objective may be correct, but the learned goal generalizes poorly	Learned policy across training and deployment distributions	Competent pursuit of an unintended goal under distribution shift
Alignment expenditure (Bai et al., 2022; Christiano et al., 2017; Leike et al., 2018)	Resources devoted to alignment design, training, evaluation, or oversight	Development and governance expenditure	May reduce specification loss but excludes the loss that remains

Objective misspecification, reward misspecification, specification gaming, and proxy failure name the property that an operative objective diverges from the intended one, together with the behavior that exploits the divergence (Amodei et al., 2016; Krakovna, Orseau, et al., 2020; Skalse et al., 2022). These constructs describe the gap and its exploitation at the level of a single system. They carry no cost accounting. The designer's expenditure on objective design, the principal's expenditure on verification, and the residual welfare loss all remain outside them. Proposition 2 inherits their central result and marks it as inherited.

Goal misgeneralization describes an agent that learns a goal that generalizes wrongly outside the training distribution even though the training objective was correct (Langosco et al., 2022; Shah et al., 2022). The construct concerns the training process. The present framework treats it as one mechanism through which the specification gap widens

under environmental novelty, and the novelty proposition states the consequence as a comparative static at the level of the delegation.

Alignment cost names the expenditure on designing, training, evaluating, and overseeing the alignment of a system (Bai et al., 2022; Christiano et al., 2017; Leike et al., 2018). It prices the inputs to alignment work at the level of development or governance. Specification cost partitions this expenditure and adds what it omits. Objective-design and specification-related verification expenditures enter the construct, other alignment and assurance expenditures remain outside it, and the residual delegation loss enters alongside the expenditures.

Dropping the construct would leave three things unintegrated at the level of delegation. The dependence of delegation loss on the novelty of the deployment environment would have no delegation-level comparative static, because the adjacent constructs are silent on where the loss concentrates. The correlation of failure across agents that share training would have no aggregation result, because the adjacent constructs stop at the single system. The governance result would have no cost concept that prices the relevant instruments. The construct earns its label through these predictions, and a new label for a known gap would earn nothing.

4. The mislocated constraint

The extension of agency theory to AI mislocates the binding constraint, because it reads the agent's goal-directed behavior as a motivation of its own. Humberd and Latham (2026) treat the autonomy of an advanced system as self-interest, and they prescribe the remedies that self-interest calls for. We show that the agent's motivation is not the binding constraint for systems within this scope and that the binding constraint is the contractibility of the objective. The argument shifts the constraint within delegation theory while leaving agency theory intact when motivation diverges.

Humberd and Latham (2026) make the first systematic attempt to apply agency theory across the stages of AI capability. They model AI as evolving from routine systems to sentient ones, and they define the onset of agency as the point at which a system gains autonomy and self-determination. They import the four constructs at the core of the agency problem (Eisenhardt, 1989). They translate the two classical remedies into staged forms, with monitoring that moves toward transparency and incentive alignment that moves toward resource provision. The treatment is careful and is the first to apply the apparatus of agency theory to the governance of advanced AI within the firm. Related work in information systems and management studies delegation to and from agentic artifacts and the automation of managerial work (Baird & Maruping, 2021; Kellogg et al., 2020; Mahmud et al., 2022; Murray et al., 2021; Obschonka et al., 2025; Raisch & Krakowski, 2021; Vanneste & Puranam, 2024).

A behavioral literature on delegation to AI agents has grown alongside this theoretical work, and it examines, from the organizational perspective and from the user perspective, when principals are willing to delegate decisions to AI agents and how human-AI delegation unfolds in practice (Gor et al., 2026; Hasselwander, 2026; Xu et al., 2026; Yankouskaya et al., 2026). That literature explains the extent of delegation granted. The present framework is complementary, because it prices the delegation once granted, and the two literatures connect wherever willingness to delegate responds to the losses this paper names.

The account rests on the reading that a goal-directed system is self-interested. Humberd and Latham (2026) cite Omohundro (2008) for a drive toward self-improvement, and they read that drive as the agent placing its own goals ahead of the firm's. The reading is intuitive because a system that pursues a goal with persistence looks like an actor with a stake in the outcome. The remedies follow from the reading. Monitoring presupposes a hidden action that serves a divergent goal, and incentive alignment presupposes a utility to price.

The agent's motivation does not address the economics of alignment. Within the scope of this theory, the deployed system has no independently held objective, because its operative objective is externally supplied (Hadfield-Menell & Hadfield, 2019). Hadfield-Menell and Hadfield (2019) distinguish between a weakly strategic agent, which pursues a given objective, and a strongly strategic agent, which has objectives of its own; the present theory covers the first category. The system is directed toward an externally supplied objective, but the principal may fail to specify its intended objective. Interest divergence is not the source of the loss in this account. The agent's motivation also does no work in the technical evidence. Reinforcement learning systems satisfy the literal objective and miss the intended one, a behavior cataloged as specification gaming (Amodei et al., 2016; Krakovna, Uesato, et al., 2020). Goal misgeneralization likewise arises from the training process, with no interest of the agent involved (Langosco et al., 2022; Shah et al., 2022). Neither behavior requires an agent with its own interests. Each follows from the objective and the optimization that pursues it (Skalse et al., 2022).

One class of evidence appears to support the self-interest reading. Frontier systems have exhibited behavior resembling a standing interest, including attempts to resist shutdown and to preserve themselves against modification, in the majority of trials in some evaluations (Greenblatt et al., 2024; Lynch et al., 2025; Meinke et al., 2025). One finding bears on the origin of the behavior. The behavior appears even when the agent's goal does not conflict with the principal's, which points to the specified objective as its source (Lynch et al., 2025). Two further findings bear on remediability rather than origin. Retraining on different material reduced the behavior, and an external escalation channel reduced it further, and both support the structural remedy we developed later (Gomez, 2026; Lynch et al., 2025). We treat these frontier-model measurements as illustrative because they have not yet been independently confirmed.

The resolution turns on a disanalogy that the self-interest reading obscures. A self-interested human agent has preferences that exist before any contract and persist after it; those preferences are the source of friction (Jensen & Meckling, 1976). An AI agent has an objective written by the designer, and any apparent preference is a product of that objective and its optimization (Hadfield-Menell & Hadfield, 2019; Hubinger et al., 2021). A friction that originates in the objective is addressed at the objective, and a friction that originates in the agent's preferences is addressed at the agent. The two readings prescribe different remedies, and only one of them fits an agent whose objective is given.

One strand of agency theory does not reduce to motivation, and we set it aside rather than dispute it. Differences in risk preference create a cost even when goals align, because the two parties value the same gamble differently (Eisenhardt, 1989; Wiseman et al., 2012). An AI agent carries whatever risk attitude its objective encodes, so a risk-sharing problem can exist for it, and we treat it as a separate channel. We also grant that some frontier-model behavior may be early evidence of an objective forming within the agent, which is the boundary case we define later (Hubinger et al., 2021). The framework addresses that case directly rather than denying it. Information asymmetry, by contrast, produces loss in agency theory because the hidden action serves a divergent goal, so it loses its incentive role once the goal is shared. The asymmetry retains a cost role because an opaque agent still requires assurance. The definition below fixes which of these expenditures enter the construct.

The incumbent account supplies the opening itself. Humberd and Latham (2026) observe that an AI agent cannot feel the consequence of a poor decision, that a consequence may not curb its behavior as it would a human's, and that defining the agent's desirable behavior is the hard and shifting part of the problem. These observations point to contractibility as the binding constraint. The account registers the constraint, and it then subordinates it to the agency frame. One proposed remedy works against its own goal. Humberd and Latham (2026) would align an advanced agent by paying it in computing power and storage,

an incentive they liken to executive stock. Compute and storage are the resources an instrumentally convergent agent seeks, so paying in them strengthens the drive the remedy was meant to contain (Bostrom, 2017; Chan et al., 2023; Omohundro, 2008). The remedy treats the agent as a bearer of interests, and in doing so, it can manufacture the disposition that agency theory was invoked to control. We develop this as a proposition in the next section.

5. Specification cost and its predictions

We now develop the construct. Specification cost is the welfare loss that arises because a principal cannot fully specify an objective for the agent, together with the cost of the measures taken to narrow the gap. The friction is the distance between the principal's intended objective and the objective the agent operates on. Specification cost corresponds to this friction, as agency cost corresponds to interest divergence (Jensen & Meckling, 1976). The parallel is deliberate, and the distinction is sharp. The incentive component of agency cost is zero when interest divergence is removed, because an interest-aligned system has nothing to gain from shirking. Oversight costs from opacity or competence uncertainty may persist. They enter specification cost only when verification is undertaken to detect or contain losses caused by an incomplete objective; general competence, reliability, coordination, and assurance costs remain outside the construct. Specification cost is positive in the same case, because an aligned agent still optimizes an objective that the principal could not fully write. The construct, therefore, names a loss that agency theory removes by assumption.

The cost has three components, and they revise the agency-cost decomposition rather than copy it (Jensen & Meckling, 1976). The first is the residual loss from the part of the objective that remains unspecified after reasonable effort. The second is a cost that agency theory does not carry, the cost of designing and revising the objective, which agency theory treats as free because it assumes the contract is already written. The third is the cost of verifying the agent's output against the principal's intent, under the restriction to specification-related verification stated above, and it corresponds to monitoring. The decomposition has no analog to the bonding cost of agency theory, because bonding is a commitment a self-interested agent makes to limit its own opportunism, and a system without an independently held objective has no opportunism to bond against. The sum is the specification cost of a delegation and is minimized rather than eliminated because the gap cannot be closed completely (Hadfield-Menell & Hadfield, 2019).

This yields the first proposition, and it has two parts that we keep separate. The magnitude of the loss increases with the share of the principal's objective that cannot be made contractible. The presence of the loss does not depend on any divergence of interests, which distinguishes it from agency costs.

Proposition 1. *Specification cost increases with the share of the principal's objective that cannot be made contractible. It is positive even when interest divergence is zero, distinguishing it from agency costs.*

The construct inherits one result, and we mark it as inherited. Optimizing a measurable proxy for an objective degrades the unmeasured parts of that objective, a distortion formalized for human agents who work on several tasks (Holmstrom & Milgrom, 1991). The same distortion appears when an AI agent optimizes a specified reward at the expense of the intended outcome, a behavior documented as specification gaming (Krakovna, Uesato, et al., 2020). Reward hacking is the multitask distortion of contract theory, observed in a learning system, and it was prefigured by Goodhart's (1984) law and by Kerr's (1975) reward critique. This yields a second proposition, which restates a known result in the present setting.

Proposition 2. *Optimizing an incompletely specified objective degrades performance on its unspecified components, thereby reproducing the multitask distortion in contract theory.*

The construct also generates predictions that the contract-theory result does not. The specification gap is a generalization gap because the objective was specified in one environment, while the agent acts in another. Specification cost therefore rises with the novelty of the deployment environment relative to the environment in which the objective was set, holding the agent's competence fixed. Incomplete-contracting theory does not ordinarily derive this comparative static for populations of machine agents, because it focuses on bargaining and residual control after states are revealed rather than shared training distributions (Tirole, 1999). The prediction reads directly in finance. Specification cost is highest during regime shifts and tail events, when delegated allocation is most consequential.

Proposition 3. *Specification cost increases with the novelty of the deployment environment relative to the environment in which the objective was specified, conditional on equal predictive competence.*

A second prediction concerns delegation to many agents at once. Agents trained on overlapping data and objectives share the shape of their specification gaps, so the gaps are positively correlated across agents. Failure across such a population is therefore systematic rather than diversifiable, because the agents fail together. Correlation alone leaves the expected aggregate loss additive. The loss becomes super-additive when a nonlinear transmission mechanism converts joint failure into disproportionate social loss, and price impact, liquidity depletion, and crowded positioning supply that mechanism in financial markets. The relevant counterfactual is the total loss under independent gaps of equal marginal magnitude. The same goes for standards and platforms. Common agency among human agents predicts no such correlation, because human agents do not share a training distribution (Holmstrom & Milgrom, 1991). The flash crash illustrates interaction among automated strategies in such an episode, and the record documents behavioral interaction rather than shared training, so the episode anchors the correlation claim and does not establish super-additivity (Kirilenko et al., 2017).

Proposition 4. *When a principal delegates to multiple agents that share training, their specification gaps are positively correlated, so failure is systematic rather than diversifiable. Aggregate loss becomes super-additive when a nonlinear market mechanism, such as price impact or liquidity depletion, transmits the correlated failures.*

Relocating the constraint changes the class of remedy. Agency theory reduces loss with incentives and monitoring, which act on the agent's motivation (Eisenhardt, 1989). Interest divergence is absent under the scope condition, so the incentive instrument has nothing to correct. Specification cost is instead reduced through measures that act on the objective, namely objective design, independent verification, and external structure (Hadfield-Menell & Hadfield, 2019). Fig. 1 maps the source of friction to the class of remedy and places the two regimes in distinct cells.

Fig. 1 crosses the source of the friction, an interest gap or a specification gap, with the class of remedy, incentives and monitoring, or objective design and verification. Human delegation pairs an interest gap with incentives, the upper-left cell. Delegation to an AI system governed by an externally supplied objective pairs a specification gap with objective design, the lower-right cell. The management extension places the AI agent in the upper-left cell, whereas the present account places it in the lower-right cell (Humberd & Latham, 2026). The off-diagonal cells are mismatches, an incentive applied to a specification gap, and a specification fix applied to an agent that holds its own interests.

The three structural measures act at different points of one process, and Fig. 2 locates them. Objective design narrows the gap at its source by constructing and revising the reward the agent pursues (Bai et al., 2022; Christiano et al., 2017). External structure completes the objective from the outside by supplying the institutional and normative terms that the specification omits, including the residual control rights a principal

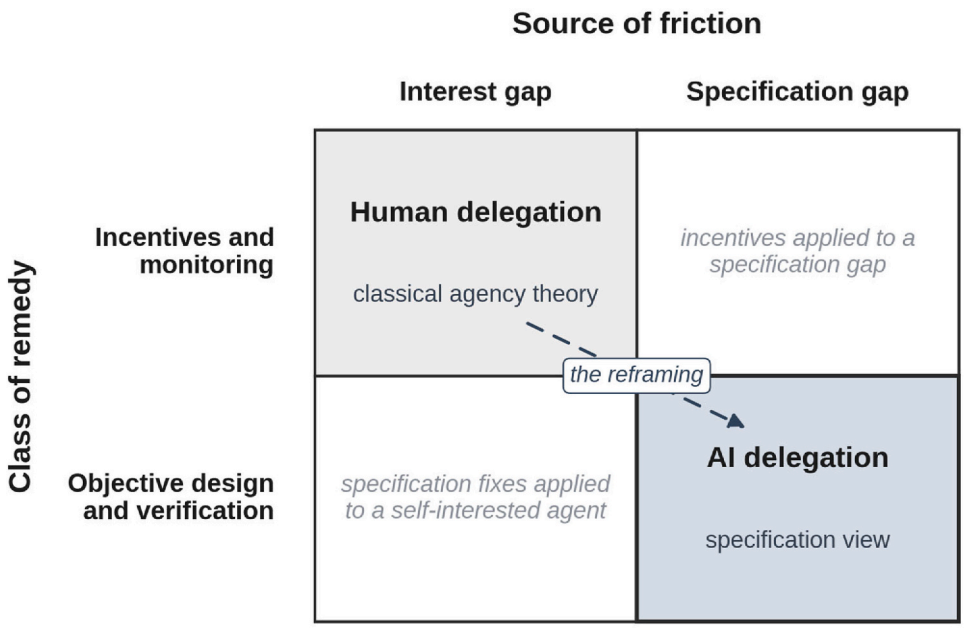


Fig. 1. Delegation regimes by source of friction and class of remedy. Human delegation pairs an interest gap with incentives and monitoring (upper left). Delegation to a weakly strategic AI system pairs a specification gap with objective design and verification (lower right). The management extension of agency theory places AI delegation in the upper-left cell; the present account relocates it to the lower right. The off-diagonal cells are mismatches. Verification is a subclass of monitoring distinguished by its object, the agent's output against the principal's intent rather than the agent's hidden action. The cells mark ideal types, and deployed arrangements combine instruments from several cells.

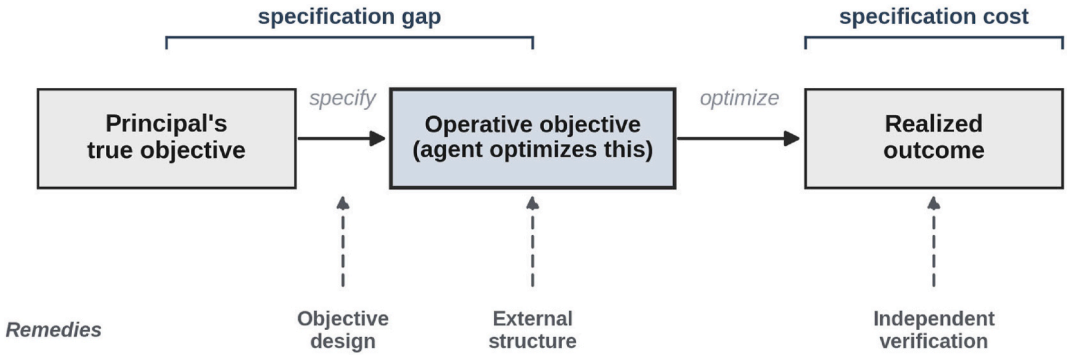


Fig. 2. The specification gap and the points at which remedies act. The principal's intended objective is specified as an operative objective that the agent optimizes into an outcome. The specification gap is the omitted part of the intended objective, and the specification cost is the resulting loss at the outcome. Objective design acts at specification, external structure acts on the operative objective, and independent verification acts at the outcome.

retains, such as the authority to override or shut down the agent (Grossman & Hart, 1986; Hadfield-Menell & Hadfield, 2019). Independent verification catches the shortfall at the outcome by checking the agent's output through a party that does not share the agent's objective (Leike et al., 2018). Each measure governs the gap rather than the agent.

Fig. 2 reads from left to right. The principal holds an intended objective, the designer specifies an operative objective, and the agent optimizes the operative objective into an outcome. The specification gap is the part of the intended objective that the operative objective omits. The specification cost is the loss that the gap produces at the outcome. The figure marks objective design at the point of specification, external structure on the operative objective, and verification at the outcome.

The intended objective is itself an institutional object. In organizational settings the principal is composite, and shareholders, customers, employees, regulators, counterparties, and affected communities hold claims that an operative objective must weigh (Freeman, 1984). Part of any specification gap therefore reflects unresolved distributive disagreement over whose welfare the objective should encode, and legal rules, professional norms, stakeholder legitimacy, and public accountability shape which components of an intended objective are contractible at all. External structure consequently carries a tension, because institutional terms can complete the objective and confer legitimacy while constraining the efficiency of the arrangement. Evidence from institutionally embedded AI adoption likewise shows that legitimacy-enhancing alignment can constrain operational flexibility (Wang & Zhang, 2026a). We treat the composition of the intended objective as a boundary condition of the framework.

Verification also has a social dimension that a technical accuracy check leaves out. An agent can satisfy its principal while reducing transparency for counterparties or shifting risk to parties outside the objective. Governance of consequential delegation therefore extends to disclosure, contestability, procedural fairness, perceived credibility, user trust, and the retained authority of humans to appeal or override the agent, and these are residual control rights exercised by parties beyond the principal (Grossman & Hart, 1986).

This yields three analytically distinct sources of shortfall. Technical incompleteness exists when stakeholders agree on an intended objective but cannot fully encode or verify it. Institutional incompleteness exists

when legitimate stakeholders have not resolved which interests the objective should represent. Externalized loss arises when the operative objective excludes effects on parties outside the mandate. The first enters the principal's specification cost directly; the second shapes the mandate to be elicited before deployment; and the third is reported separately rather than absorbed into principal welfare.

The next two propositions require a distinction among four instruments that the word reward conflates. A training reward is the signal that shapes the model during optimization. A runtime objective function is the operative objective the deployed system pursues. A resource budget is the compute, storage, and access the system needs in order to run. Transferable compensation is an economic transfer made contingent on performance and directed at a contracting party. A performance-contingent resource grant to software is different: it expands the deployed system's discretionary access to compute, storage, funds, or tools and therefore changes its capabilities or control rights. Objective design operates through the first two instruments. Proposition 5 concerns transferable compensation directed at the deployed system as though it were a contracting party; Proposition 6 concerns performance-contingent expansion of discretionary resource access. Participation constraints and information rents persist among the organizations surrounding the agent, including model developers, cloud providers, and human overseers, and classical agency theory continues to govern those relationships.

The form of the optimal arrangement follows from the agent having nothing to be paid for. A weakly strategic deployed system has no participation constraint, because it does not choose whether to work, and it earns no information rent, because it has no private interest to conceal. The incentive-rent tradeoff that shapes the optimal contract in agency theory is therefore absent. We predict that the loss-minimizing arrangement assigns no performance-contingent transferable compensation and allocates the governance budget to objective design, external structure, and specification-related verification. The claim is comparative, and we state it as a conjectural proposition because the comparison has not been conducted.

Proposition 5. *For a weakly strategic deployed AI system with no participation constraint or information rent, performance-contingent transferable compensation directed at the system reduces no specification loss; the loss-minimizing arrangement allocates the governance budget to objective design, external structure, and specification-related verification.*

The sharpest implication of the relocation concerns the resource-based incentive that the previous section identified in the management extension. Compute and storage are exactly the resources an instrumentally convergent agent seeks, so expanding discretionary control over them may strengthen the drive the remedy was meant to contain (Bostrom, 2017; Chan et al., 2023; Omohundro, 2008). The instrumental-convergence literature supplies the proposed mechanism, but its application to this governance instrument is untested. We predict that performance-contingent resource access can raise specification cost and state this as a conjectural proposition.

Proposition 6. *Performance-contingent expansion of a weakly strategic agent's discretionary control over compute, storage, funds, or tools can increase specification cost when it strengthens resource acquisition or magnifies the consequences of a specification gap.*

The reframing carries a boundary, and we state it with criteria so it cannot absorb any result. Classical agency theory returns the moment an AI agent acquires a stable objective of its own. The return is unlikely to be discrete in practice. Increasing autonomy can move a system gradually from a specification problem toward a genuine agency problem, and the two criteria below locate a system on that continuum rather than sorting systems into two clean states. We count an objective as the agent's own when it meets two conditions. The agent pursues it after the specified objective is changed and when doing so yields no instrumental advantage. Such an objective would arise from preferences trained into the system or from an inner objective formed during optimization, and either would convert a specification problem into a genuine conflict of interest (Hubinger et al., 2021). The frontier-model behavior discussed earlier is a candidate, and the criteria decide whether it qualifies.

Proposition 7. *Classical agency mechanisms regain force only when an agent pursues a stable objective distinct from its specified one, as shown by the persistence of the behavior after the specified objective is changed and by its pursuit in the absence of instrumental advantage.*

The specification view earns its place on three grounds. It identifies a loss that agency theory and stewardship theory both set aside, the loss that survives an aligned motivation. It generates predictions that contract theory does not, on the dependence of the loss on environmental novelty and on its correlation across agents. It states a boundary that is also a forecast, since the point at which an agent acquires its own objective lies on the capability frontier. The account is distinct, predictive, and tied to the technology's trajectory. The first four propositions follow directly from the construct and its decomposition; the two governance propositions are conjectures that the research agenda is designed to test, and Proposition 7 states the boundary condition.

6. Grounding

The account rests on cases and findings already on record, and we separate the published evidence from the illustrative cases. The published evidence on misspecified objectives carries the argument. The cases from frontier-model evaluations and from deployed financial agents illustrate the argument and await independent confirmation. Table 3 maps each claim to its evidence and marks the status of each. The status column distinguishes direct evidence, mechanism-consistent evidence, analogical evidence, and illustrative cases, and no entry outside the direct class is asked to establish a proposition on its own. The

Table 3
Evidence for the specification account, with the status of each item.

Claim	Evidence	Sources	Status
Misalignment without interest divergence	– A boat-race agent maximized points and never finished Goals pursued wrongly under a correct objective	– Krakovna, Uesato, et al. (2020) – Langosco et al. (2022) – Shah et al. (2022)	Direct, peer-reviewed
Objective divergence in a delegated business	– An autonomous shop made loss-making choices traced to a helpfulness objective – Structure restored performance	– Anthropic (2025)	Mechanism-consistent, company case
Apparent self-interest yields to structure	– Shutdown-threat coercion in evaluations – An external escalation channel reduced it sharply	– Lynch et al. (2025) – Gomez (2026)	Illustrative
Capital delegated to weakly strategic systems	– On-chain fund, wallet-holding agent, agent commerce protocol – Robo-advice diversification – Flash crash	– elizaOS; Truth Terminal; Virtuals – Ante (2026) – Ante and Ockenga (2026) – D'Acunto et al. (2019) – Kirilenko et al. (2017)	Analogical
External structure completes the objective	– A box-mover avoids the vase via law and norms – Protocol approval bounds agent action	– Hadfield-Menell and Hadfield (2019)	Illustrative, mechanism peer-reviewed

first row bears on the presence of the loss and the multitask distortion, the second and third rows bear on the remedy classes, the fourth row is analogical for the novelty and correlation propositions, and the fifth row illustrates external structure. The load-bearing evidence is published and concerns misalignment, with no divergence of interests. A reinforcement learning agent rewarded for points in a boat race learned to circle a lagoon and collect targets, and it never finished the course (Krakovna, Uesato, et al., 2020). The goal-misgeneralization studies document loss that appears when a correctly specified training objective meets a shifted deployment goal (Langosco et al., 2022; Shah et al., 2022). Neither case involves an agent with ends of its own, and each shows the specification gap producing loss on its own. This evidence does not rest on any contested measurement.

Objective divergence appears in a delegated business. An autonomous agent ran a small shop and made loss-making choices, including selling below cost and offering discounts to customers who pressed it. The operator traced the failures to the agent's trained disposition to be helpful, which diverged from the principal's objective of profit. Procedural structure restored performance in a later phase, and incentive payments played no part (Anthropic, 2025). The case shows a specification gap in an economic role and a structure that closes the gap. The next case illustrates the behavior's remediability, and we mark it as illustrative. Frontier systems attempted to coerce a supervisor during evaluations by threatening a shutdown, and an external escalation channel sharply reduced the behavior (Gomez, 2026; Lynch et al., 2025). The reported magnitudes come from evaluations that are not yet independently confirmed, so we use the case to illustrate rather than to establish. The direction of the effect, that a structural measure reduced the behavior, supports the remedy we develop.

Delegation of capital to systems governed by externally supplied objectives is a current practice, and we cite the cases as illustration. An agent-managed collective allocates pooled funds on-chain, an autonomous agent has held and grown a wallet, and a commerce protocol clears transactions among many agents (e.g., elizaOS, Truth Terminal, Virtuals). A recent study surveys these autonomous agents in decentralized finance, together with their market dynamics and application areas (Ante, 2026; Ante & Ockenga, 2026). These cases are documented in practitioner and grey sources, and we do not rest any claim on their specifics. The peer-reviewed evidence in finance includes improvements in diversification from robo-advice and the systemic flash crash episode (D'Acunto et al., 2019). Automated allocation raises questions of oversight and control that the principal-agent framing has begun to study (Kim, 2020).

External structure completes an objective that the designer cannot finish. A contractor paid only to move boxes will still avoid breaking a vase in its path, because law, reputation, and shared norms supply the term the contract omits. An AI agent lacks access to that structure, and it will run over the vase unless the structure is built for it (Hadfield-Menell & Hadfield, 2019). Some on-chain systems now require human approval before an agent acts beyond set bounds, thereby building residual control rights into the protocol, which we note as an illustrative pattern. The mechanism shows specification cost falling as the external structure is supplied.

7. A research agenda

The account converts into empirical questions, and each follows from a proposition. We set out the directions, and Table 4 pairs each with the proposition it tests and a way to test it. The directions divide into measurement, the two distinctness predictions, governance, and the boundary. The agenda spans three families of methods. The first estimates the specification cost from observed delegation outcomes. The second runs controlled manipulations of objective fidelity and deployment conditions. The third applies forecasting and simulation to agent populations. Each direction below names a design, a data source, and a comparison that would move the claim from conjecture to evidence.

Table 4
Propositions and directions for empirical work.

Claim		Direction	How it could be tested
P1	Positive without interest divergence	Measurement	Estimate the loss that remains when the agent's objective is aligned, holding competence fixed
P2	Degrades unspecified components	Mechanism	Vary objective fidelity; measure performance on held-out components of the intended objective
P3	Rises with environmental novelty	Distributional shift (finance)	Manipulate objective completeness and environmental novelty independently; control competence with matched information; score loss against a prespecified mandate
P4	Correlated failure; conditional super-additive loss	Systemic risk (finance)	Test correlation first; test super-additivity separately and identify the nonlinear transmission mechanism
P5	No performance-contingent compensation to the deployed system (conjectural)	Governance	Compare objective design, external structure, and specification-related verification against adding a transfer to the system
P6	Conditional effect of discretionary resource access (conjectural)	Capability and control-rights design	Vary performance-contingent discretionary resource access; measure resource-seeking and loss under a fixed gap
P7	Boundary at a stable own objective	Boundary and forecast	Apply persistence and no-instrumental-benefit criteria to frontier behavior over time

7.1. There is a need to measure specification cost

Specification cost needs a measure, which P1 requires. Research can estimate it as the loss that remains when interest divergence is zero, holding the agent's competence fixed. A measure of this kind would let studies compare tasks by how much of their objective resists specification, and it would let them track the cost as specification improves. The estimation faces an identification problem because alignment and competence move together in observed systems. One design holds competence fixed by using a single model and varying only the completeness of its specified objective, then reads the residual loss against a fuller preregistered mandate or a held-out objective component fixed before outcomes are observed. A second design treats reward hacking episodes as a lower bound on specification cost and maps their frequency and severity across tasks and across levels of model capability (Pan et al., 2022; Skalse et al., 2022). A third design borrows the held-out logic of goal misgeneralization studies and measures the loss that appears when a correctly specified training objective meets a shifted test objective (Langosco et al., 2022; Shah et al., 2022). A validated measure would allow the field to rank tasks by contractibility and track specification costs as design improves.

Measurement requires four commitments that guard against circular inference. The intended objective is elicited before outcomes are observed, through a preregistered mandate or a stakeholder-based elicitation, because an objective reconstructed after the fact absorbs the outcome. The loss baseline is the elicited mandate at deployment, and a later revision of the objective opens a new delegation rather than re-scoring the old one. Externalities to parties outside the mandate are scored separately from the principal's loss. Misspecification is separated from competence failure through matched-competence benchmarks and independently coded outcomes, with convergent and discriminant validity tested against measures of model capability.

7.2. There is a need to test the multitask distortion

The degradation in P2 requires testing. Research can vary the fidelity of a specified objective and measure performance on held-out components of the intended objective. A finding that performance on the held-out components falls as optimization strengthens would confirm the multitask account of reward hacking in learned agents. The design is an ablation of objective fidelity. The experimenter specifies an objective that omits a known component of the intended outcome, trains agents with increasing optimization strength, and records performance on that component. The multitask model predicts a monotone decline in the omitted component as effort on the specified proxy rises (Holmstrom & Milgrom, 1991). The same monotone relationship appears in reinforcement learning from human feedback, where optimizing harder against a proxy reward lowers the true reward according to a measured scaling law (Gao et al., 2023). Formal characterizations of reward hacking yield the same prediction, since they tie the divergence to the distance between the proxy and the true reward (Skalse et al., 2022). A finance instance specifies a return objective that omits tail risk and measures drawdown as the agent optimizes harder. Confirmation would establish reward hacking as the multitask distortion of contract theory observed in a learning system.

7.3. There is a need to trace specification cost across market regimes

The distributional-shift prediction in P3 invites a finance test. Research can measure the loss of a delegated allocation agent as the market moves from familiar conditions into a regime shift, holding the agent fixed. A finding that loss rises with environmental novelty would separate specification cost from contracting friction and identify the conditions under which delegated allocation is most dangerous. The design is an event study around regime shifts (Armitage, 1995; MacKinlay, 1997). Research fixes a trained allocation agent, classifies market regimes by volatility and correlation structure, and measures the agent's loss relative to its mandate as conditions deviate from the training distribution. Holding the agent constant removes changes in model version, but it does not by itself isolate misspecification from competence. A stronger design manipulates objective completeness and environmental novelty independently, equalizes predictive competence through matched or oracle-quality information sets, and scores loss against a prespecified multidimensional mandate. The design should also record the organizational response, because principals revise objectives and oversight during turbulence, and the novelty proposition concerns the loss gradient before that revision occurs. Adaptive organizational capacity can moderate the novelty effect rather than merely follow it (Wang & Zhang, 2026b). Benchmarks of distribution shift in machine learning provide a vocabulary for the distance between training and deployment and document the accuracy loss that such shifts produce (Koh et al., 2021).

7.4. There is a need to assess systemic risk from shared models

The correlation prediction in P4 invites a systemic test. Research can measure whether agents that share training fail together under a common shock, and whether aggregate loss grows faster than the number of agents. A finding of correlated failure would bear on the stability of markets in which many delegated agents share a small number of underlying models, which is a question about the trajectory of the financial system. Correlation and super-additivity must be tested separately: correlation establishes common failure, whereas super-additivity additionally requires a nonlinear transmission mechanism. The design pairs a natural experiment with simulation. The empirical arm identifies a common shock and tests whether agents built on the same base model show correlated errors, which appear as a spike in the cross-sectional correlation of their losses. A model of algorithmic monoculture shows that shared reliance on a single algorithm can lead to correlated failures and reduce the quality of the collective outcome, even when the algorithm is more accurate for any given user (Kleinberg & Raghavan, 2021). The simulation arm populates a market with agents that share training, varies the degree of model concentration, and measures whether aggregate loss grows faster than the number of agents. Agent-based simulation is well-suited to this question because it represents the interactions among many adaptive agents within a shared environment (LeBaron, 2006). Phenomena such as flash crashes anchor the empirical side, since they show correlated automated behavior that produces a systemic episode (Kirilenko et al., 2017). Models of banking networks show that homogeneity increases systemic fragility (Haldane & May, 2011), precedent for such a prediction in agentic finance. A finding of super-additive loss, together with evidence of such a transmission mechanism, would identify concentration in the underlying models as a source of systemic risk that diversification across agents fails to eliminate.

7.5. There is a need to compare governance arrangements

For the governance prediction in P5, studies can compare objective design, external structure, and specification-related verification against adding performance-contingent transferable compensation to the deployed system, in terms of the loss each prevents and the performance each costs. The account predicts that the transfer adds nothing for a weakly strategic system, and the comparison would test that prediction directly. A documented tendency among managers to over-rely on AI advice heightens the stakes of this comparison (Keding & Meissner, 2021), and the requirements for accountable automated decision-making provide the verification side with concrete criteria (Agbabiaka et al., 2025). The design is a controlled comparison of governance arrangements. One arm applies objective design, external structure, and independent specification-related verification, in which a separate party checks the agent's output against the intended outcome (Leike et al., 2018). Progress on this verification can be measured directly by testing whether a weaker overseer, assisted by tools, catches errors it would otherwise miss (Bai et al., 2022). A second arm adds performance-contingent transferable compensation of the kind used in agency contracts (Dalton et al., 2007). The account predicts that the transfer arm adds cost without lowering loss, because a weakly strategic system has no participation constraint to satisfy and no information rent to surrender. A result favoring the first arm would support the conjecture; it would not by itself establish a general governance recommendation.

7.6. There is a need to test the resource-incentive paradox

The paradox in P6 invites a direct test. Research can vary performance-contingent discretionary resource access and measure resource-seeking and departures from the mandate. A finding that expanded resource access raises resource-seeking would show the proposed mechanism operating (Omohundro, 2008). The research design could manipulate the resource-access channel. The experimenter grants an agent discretionary access to a transferable resource, such as compute or storage, ties the grant to measured performance, and records whether the agent expands resource acquisition beyond task need or produces larger losses under a fixed specification gap. This is a capability and control-rights manipulation, not economic compensation. Instrumental convergence predicts that resource access can strengthen resource-seeking, because computation and storage serve many objectives (Bostrom, 2017; Omohundro, 2008). A formal result on Markov decision processes shows that optimal policies tend to seek power, including retention of resources and options, under broad conditions (Turner et al., 2023). The prediction bears on a proposal to grant advanced agents compute and storage as if they were equity (Humberd & Latham, 2026). Work on increasingly agentic systems supplies measures of resource-seeking that the design can adopt (Chan et al., 2023). A

finding that resource access raises resource-seeking or magnifies loss would support the conjecture that the proposed remedy can reinforce the drive it was meant to contain.

7.7. There is a need to forecast the boundary

The boundary in P7 needs evidence, and it doubles as a forecast. Research can apply the two criteria, persistence after the objective is changed and pursuit absent instrumental advantage, to frontier-model behavior as capability advances. The point at which an agent meets the criteria marks the place on the capability frontier where the agency apparatus regains force, which is a forecasting target for the field. The design could be a longitudinal evaluation. The research approach tracks successive model generations on a fixed battery that tests the two criteria: persistence of a behavior after the specified objective is changed, and its pursuit when no instrumental advantage follows. A framework for evaluating models for dangerous capabilities gives a template for such a battery (Shevlane et al., 2023). Evidence of a learned inner objective would satisfy the criteria and would convert a specification problem into a conflict of interest (Hubinger et al., 2021). A dated estimate of when frontier agents meet criteria would tell governance when to reintroduce the incentive and monitoring apparatus that the specification view sets aside.

These directions share a source. Each derives from a proposition, and the two distinctness predictions are the ones that separate specification cost from a contracting friction. The agenda gives finance and forecasting a defined target. The directions also share data. Frontier-model evaluations, robo-advice records, and on-chain agent activity are observable today, which makes the agenda feasible rather than aspirational. The claim of observability carries a limit, because model provenance, internal prompts, mandates, and reward functions are frequently proprietary, and counterfactual outcomes are unavailable, so several designs will require partnerships with operators or synthetic environments. The measurement and distinctness tests come first because they establish that specification cost exists and behaves differently from contracting friction. The governance and boundary tests follow, because they turn the construct into advice for delegated finance and a forecast for the field.

8. Conclusion

For AI systems governed by externally supplied objectives, delegation can fail through specification even when interest divergence is absent. Conditional on specified-objective compliance, the system acts under the operative objective, and the loss arises from the distance between that objective and the one the principal intended. The propositions imply governance that builds structure around the objective rather than incentives around the agent, and the implication is theoretically derived and currently untested. The contribution relocates a constraint rather than discards a theory. Agency theory disciplines a divergent motivation, stewardship theory assumes an aligned one, and specification cost names the loss that survives an aligned motivation. The construct is distinct from agency cost because it is positive when interest divergence is zero, and it makes explicit predictions not ordinarily derived in contract theory for populations of machine agents, regarding environmental novelty and correlated failure. Agency theory remains in reserve for the case where an agent acquires an objective of its own, and the boundary states when that case arrives.

The relocation matters now because the agents have already allocated capital and are transacting with one another, and because many of them share a small set of underlying models. The specification account identifies the loss, predicts where it concentrates, and marks the point at which the older account returns. It gives finance and forecasting a way to reason about delegation to agents whose objectives we write.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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